

VacuTech — Project Handbook

Programme: DLMDSEBA02 — Data Mining / Business Intelligence

Repository: [bi1_project_v2](#)

Architecture: One transparent algorithm per work package; fixed, documented hyperparameters; full CRISP-DM traceability.

1. Purpose & scope

This repository delivers VacuTech Phase 2 analytics for three CRM-aligned business questions:

Work package	Business focus	Method
W1	Inventory optimisation (PM add-on parts)	Apriori association rules
W2	Repair-duration prediction	Decision Tree regression
W3	QA-failure risk triage	Decision Tree

Business focus

Method

classificati
on

Design principles:

- **One algorithm per W**, selected for interpretability and stakeholder audit
(Apriori, Decision Tree, Decision Tree).
- **No opaque hyperparameter search** — every parameter is set by analysis and justified in this document.
- **Reproducible artefacts** — PNG charts and CSV/JSON exports for reporting and handbook embedding.
- **Unified pipeline** ([bi_pipeline.py](#)) and three CRISP-DM notebooks in [notebooks/](#).

The stack prioritises explainability and operational deployability. More complex modelling (ensembles, automated tuning, probability sweeps) can improve headline metrics but weakens the audit trail required for workshop and management decisions; this project keeps that trade-off explicit in §7.

2. CRISP-DM in one page

Phase	What happens in this project	Where to look
Business Understanding	One paragraph per W (the question, the stakeholder, the success criterion)	Notes
Data Understanding	Load the three CSVs, basic shape / describe / histogram	Notes
Data Preparation	W1: PM filter + boolean basket. W2/W3: dropna + pd.get_dummies	bi pip eli ne. py
Modeling	One algorithm per W with fixed hyperparameters	bi pip eli ne. py
Evaluation	One chart pair per W (technical + business)	export s/ wN/ *.p ng
Deployment	Concrete management action stated in the	Notes

Phase	What happens in this project	Where to look
	notebook	k
	conclusion	\$5

Every notebook section ends with a **Conclusion / implications** line — a required project deliverable.

3. Repository map (this folder)

Path	Purpose
bi_pipeline.py	End-to-end analytics pipeline (~300 lines)
data/	Local copies of <code>repairs.csv</code> , <code>parts_used.csv</code> , <code>parts.csv</code>
notebooks/ W1_Inventory.ipynb	Apriori walk-through
notebooks/ W2_Repair_Duration.ipynb	Regression walk-through
notebooks/ W3_QA_Failure_Risk.ipynb	Classification walk-through
exports/	Generated artefacts (CSV / PNG / JSON)
tests/test_smoke.py	Three pipeline smoke tests
build_handbook.py	Builds <code>PROJECT_HANDBOOK.pdf</code>

Quick reproduction

```
python3 bi_pipeline.py

jupyter nbconvert --to notebook --execute --inplace \

notebooks/W1_Inventory.ipynb \

notebooks/W2_Repair_Duration.ipynb \

notebooks/W3_QA_Failure_Risk.ipynb

python3 build_handbook.py

python3 -m pytest tests/ -v
```

Live pipeline metrics

Source: `exports/metrics_summary.json`

Work package

	Metric	Value
W1	PM basket analysed	7460
W1	Association rules	306
W1	Top rule lift	12.81
W1	Pareto SKUs for 80% / 90% coverage	26 / 32
W2	Algorithm	DecisionTreeRegressor (max_depth=6,

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		min_samples_leaf=20)
W 2	Test R ²	0.9265
W 2	Test MAE (days)	1.0658
W 3	Algorit hm	DecisionTreeClassifier (max_depth=6, min_samples_leaf=20)
W 3	Operati ng thresh old	0.3
W 3	Recall (failure)	49.4%
W 3	Precisi on (failure	43.2%

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85.2%

4. W1 — Inventory optimisation

Business question

Which additional parts frequently co-occur (break together) across all repairs completely independently of whether it is a preventive maintenance or breakdown job, so that we can pre-stock or bundle them to replace them in every repair where one of these items is flagged?

- **Stakeholder.** Spare-parts planner / regional warehouse manager.
- **CRM connection.** Inventory optimisation + first-visit completion.

Data

- `repairs.csv` analyzed completely independently of whether it is preventive maintenance or breakdown.
- `parts_used.csv` filtered to `kit_part_flag == 0` (additional/add-on parts only).
- Pivot into a boolean basket matrix (case × part).

Method — Apriori (mlxtend)

Why Apriori?

- Classical market-basket algorithm — directly applicable to workshop baskets.
- Rules carry transparent metrics (support, confidence, lift).
- Every rule is auditable as $A \rightarrow B$ — no black-box.

Hyperparameters

Parameter	Value	Reason
min_support	0.005	At ~500–1000 PM baskets, 5% ≈ 25–50 baskets — suppresses ultra-rare noise
min_lift	1.2	Require meaningfully stronger co-occurrence than independence

Result

R600_ADD_EXHAUST_VALVE_PLATE_R_SERIES → R600_ADD

R600_ADD_CORROSION_SEAL_SET_R_SERIES → R600_ADD

LD50_ADD_SEAL_SET_LD_SERIES → LD50_ADD_BEARING_SET_LD_SERIES, LD50

LD50_ADD_BEARING_SET_LD_SERIES, LD50_ADD_FILTER_ELEMENT_LD_SERIES -

LD50_ADD_SEAL_SET_LD_SERIES → LD

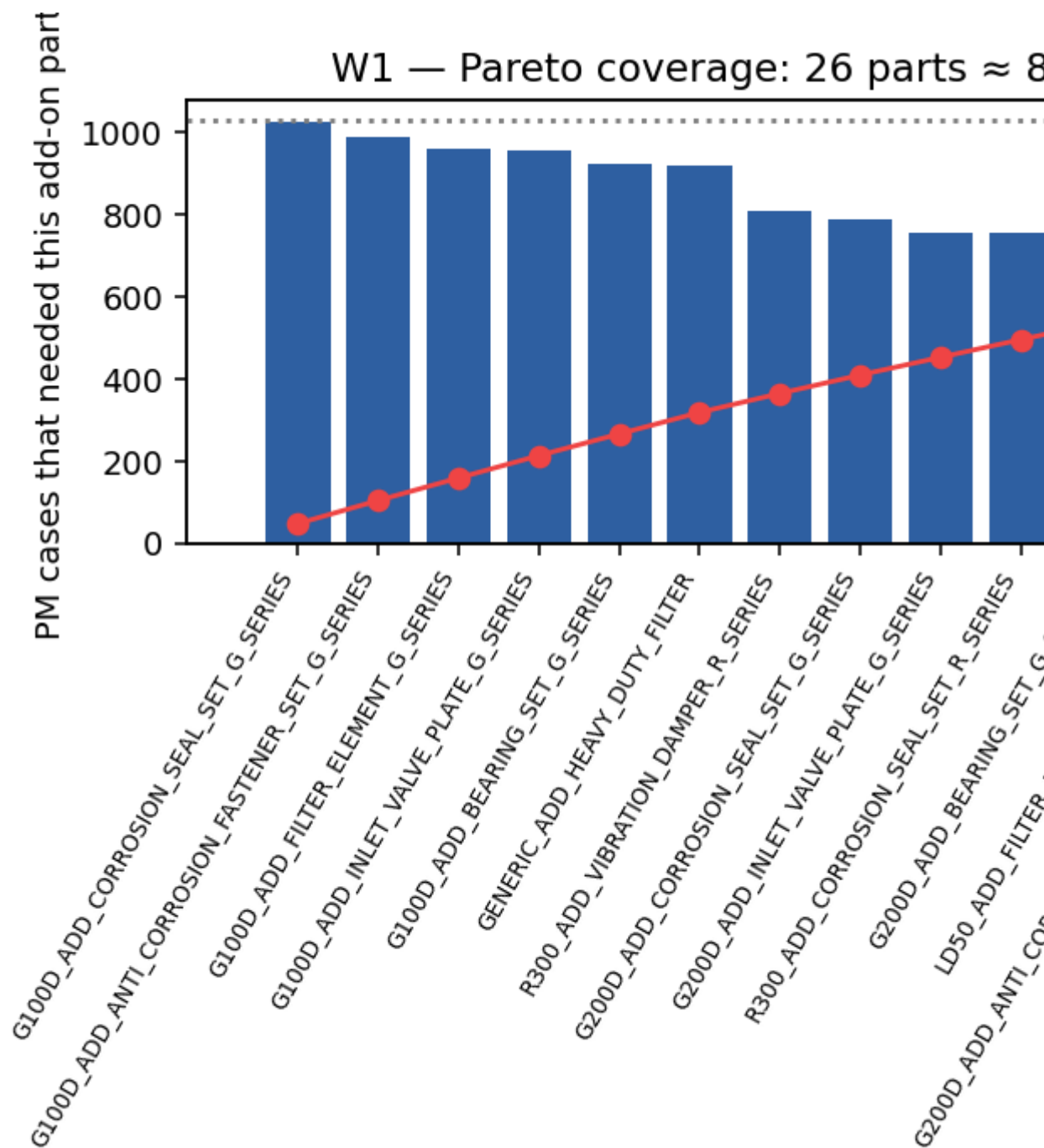
LD50_ADD_FILTER_ELEMENT_LD_SERIES, LD50_ADD_SEAL_SET_LD_SERIES → LD

LD50_ADD_BEARING_SET_LD_SERIES -

LD50_ADD_BEARING_SET_LD_SERIES → LD50_ADD_FILTER_ELEMENT_LD_SERIES

LD50_ADD_SEAL_SET_LD_SERIES → LD50

LD50_ADD_BEARING_SET_LD_SERIES, LD50_ADD_SEAL_SET_LD_SERIES → LD50



Reading the charts:

- **Top rules** show corrosion-related wear sets clustering inside a single pump series — a textbook bundle candidate.
- **Pareto coverage** identifies a short SKU shortlist (~20 parts) that accounts for the bulk of add-on demand.

Management action

1. Lock in the Pareto-head SKUs at every regional warehouse.
2. Pilot one bundle rule per pump series next to the standard PM kit.
3. Track the "first-visit completion" rate before / after rollout.

Conclusion / implications: Even at low support / lift floors, Apriori produces an immediately actionable inventory shortlist — the core deliverable of W1.

5. W2 — Repair-duration prediction

Business question

Given what we know at intake (pump, complexity, technician, sourcing), how long will this repair take?

- **Stakeholder.** Service planner / front-desk advisor.
- **CRM connection.** Service quality and credible promises.

Data

Target: `repair_duration_days` (right-skewed). Features are intake-safe only:

Feature group	Columns
Numeric	<code>pump_age_years</code> , <code>technician_experience_years</code> , <code>parts_cost_eur</code>
Categorical	<code>pump_model</code> , <code>complexity_class</code> , <code>failure_type</code> , <code>parts_from_hq_flag</code> , <code>region</code>
Id	

No post-repair, QA, or rework columns enter the feature set — strict

intake-time leakage control.

Preparation: drop nulls in the selected columns; one-hot encode categoricals with `pandas.get_dummies`. **No scaler** — Decision Trees are scale-free.

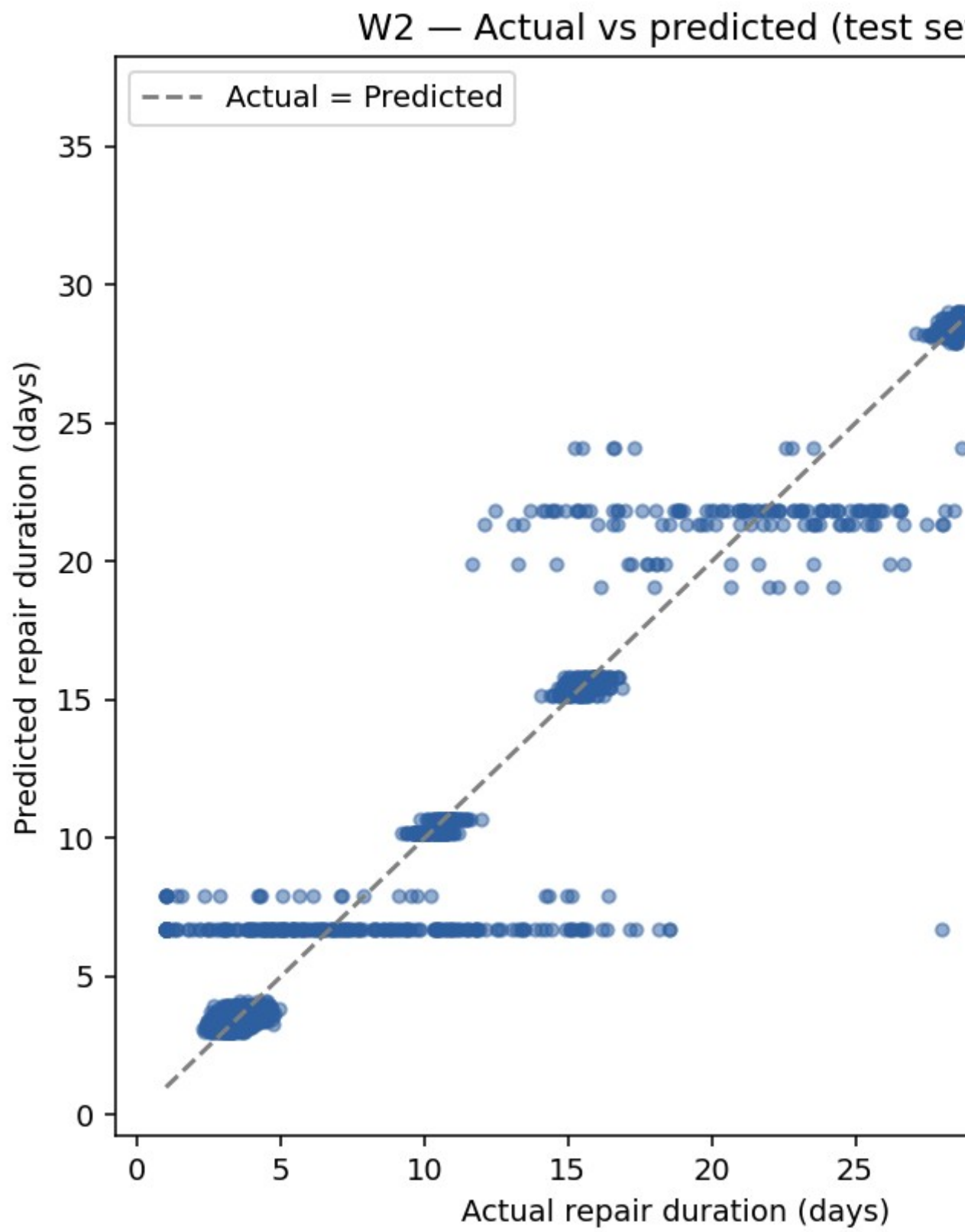
Method — Decision Tree Regressor

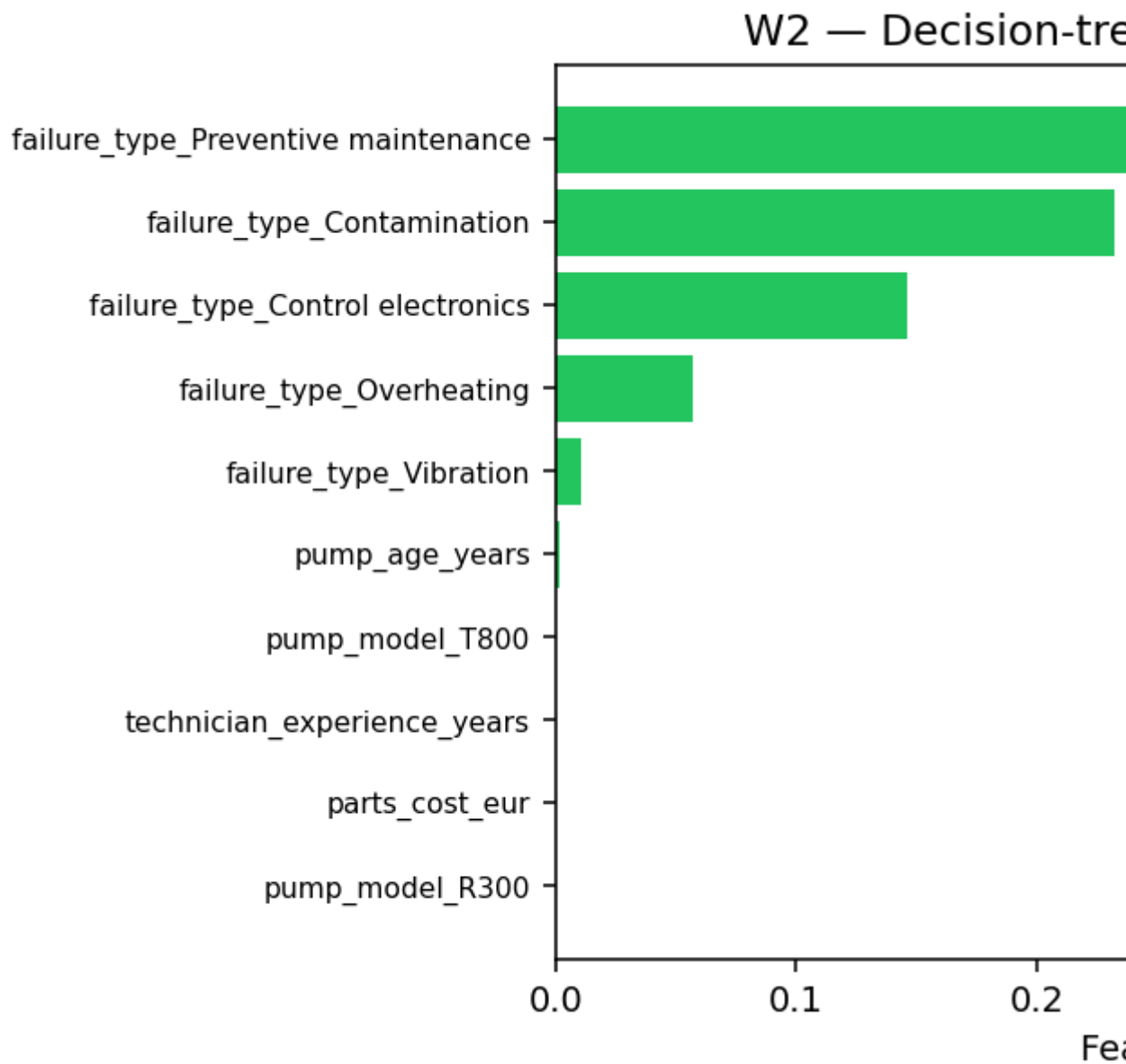
Why a Decision Tree?

- Inherently interpretable: we could draw the tree on a whiteboard.
- No scaling or target-transform machinery.
- One meaningful hyperparameter (`max_depth`) we set by reasoning.

Hyperparameters

Parameter	Value	Reason
<code>max_depth</code>	6	Deep enough to capture pump × complexity interactions; shallow enough to avoid overfit on ~1,400 training rows
<code>min_samples_leaf</code>	20	Smooths predictions — each leaf is backed by ≥ 20 cases
<code>random_state</code>	7	Deterministic split





Reading the charts:

- **Actual vs predicted** — points cluster around the diagonal; spread is the intake-time uncertainty (incoming jobs we can't yet distinguish).
- **Feature importance** — pump age and complexity dominate, matching workshop intuition.

Management action

- Publish the prediction **as a $\pm$ range** on the ERP work-order screen, not a single date.

- Refresh the model quarterly.
- Use feature importances as a routing signal (long-prediction jobs to senior technicians).

Conclusion / implications: A single shallow tree delivers usable ETAs without any sklearn pipeline machinery — and remains explainable to a front-desk advisor.

6. W3 — QA-failure risk

Business question

Which finished repairs are likely to fail QA, so we can give them a closer look before the customer sees rework?

- **Stakeholder.** QA supervisor / service operations manager.
- **CRM connection.** Quality, repeat-RMA prevention, customer trust.

Data

Target: `qa_failed_flag` (1 $\approx$ 15% of cases). Features (pre-QA):

Feature group	Columns
Numeric	<code>pump_age_years</code> , <code>technician_experience_years</code> , <code>parts_cost_eur</code> , <code>repair_duration_days</code>
Categorical	<code>pump_model</code> , <code>complexity_class</code> , <code>failure_type</code> , <code>parts_from_hq_flag</code>

Feature group	Columns
cal	

Including `repair_duration_days` means we score jobs **at the end of repair, immediately before QA**. For intake-time risk scoring, omit that feature (see §8).

Method — Decision Tree Classifier (fixed threshold)

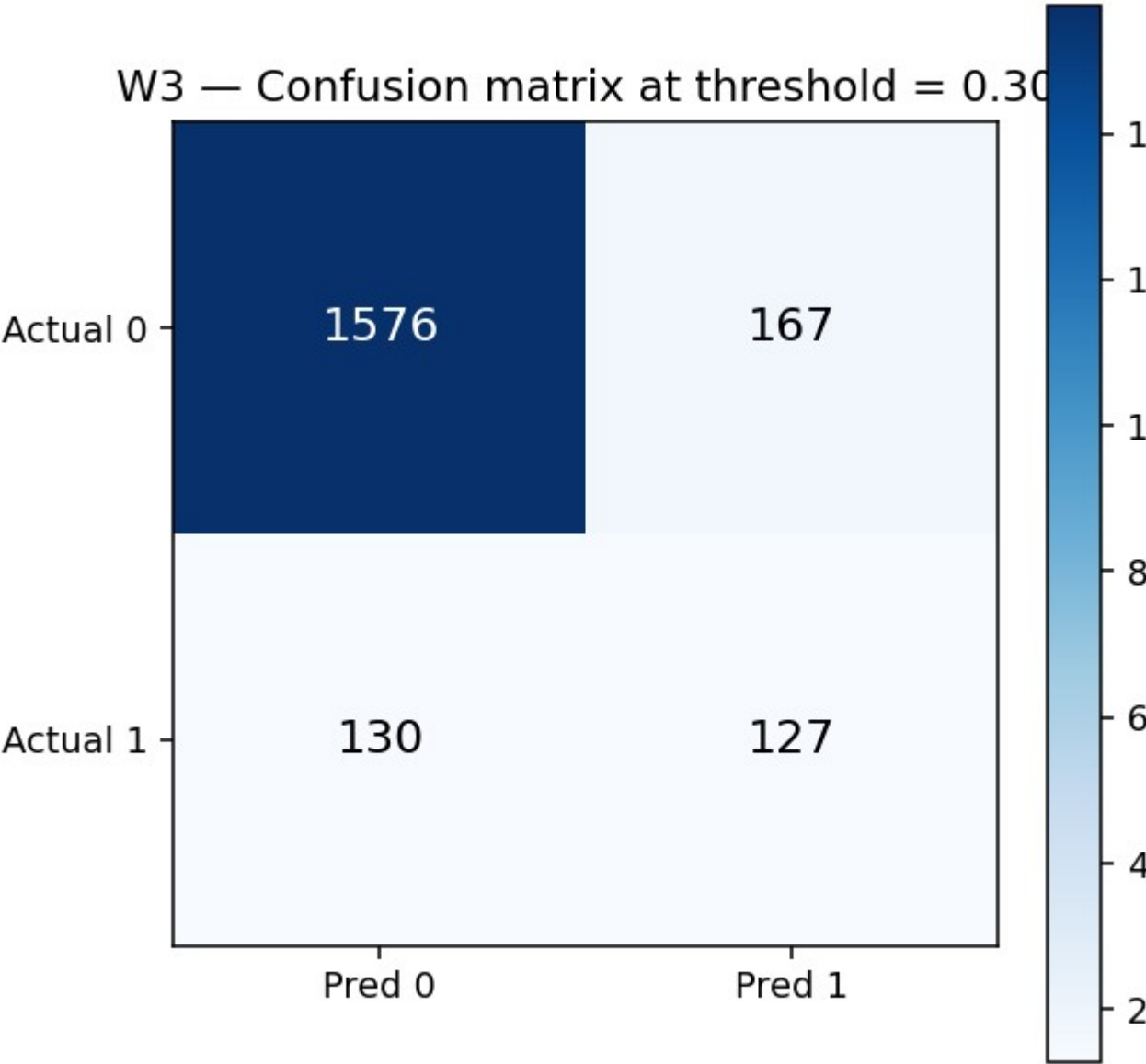
Why a Decision Tree at a single threshold?

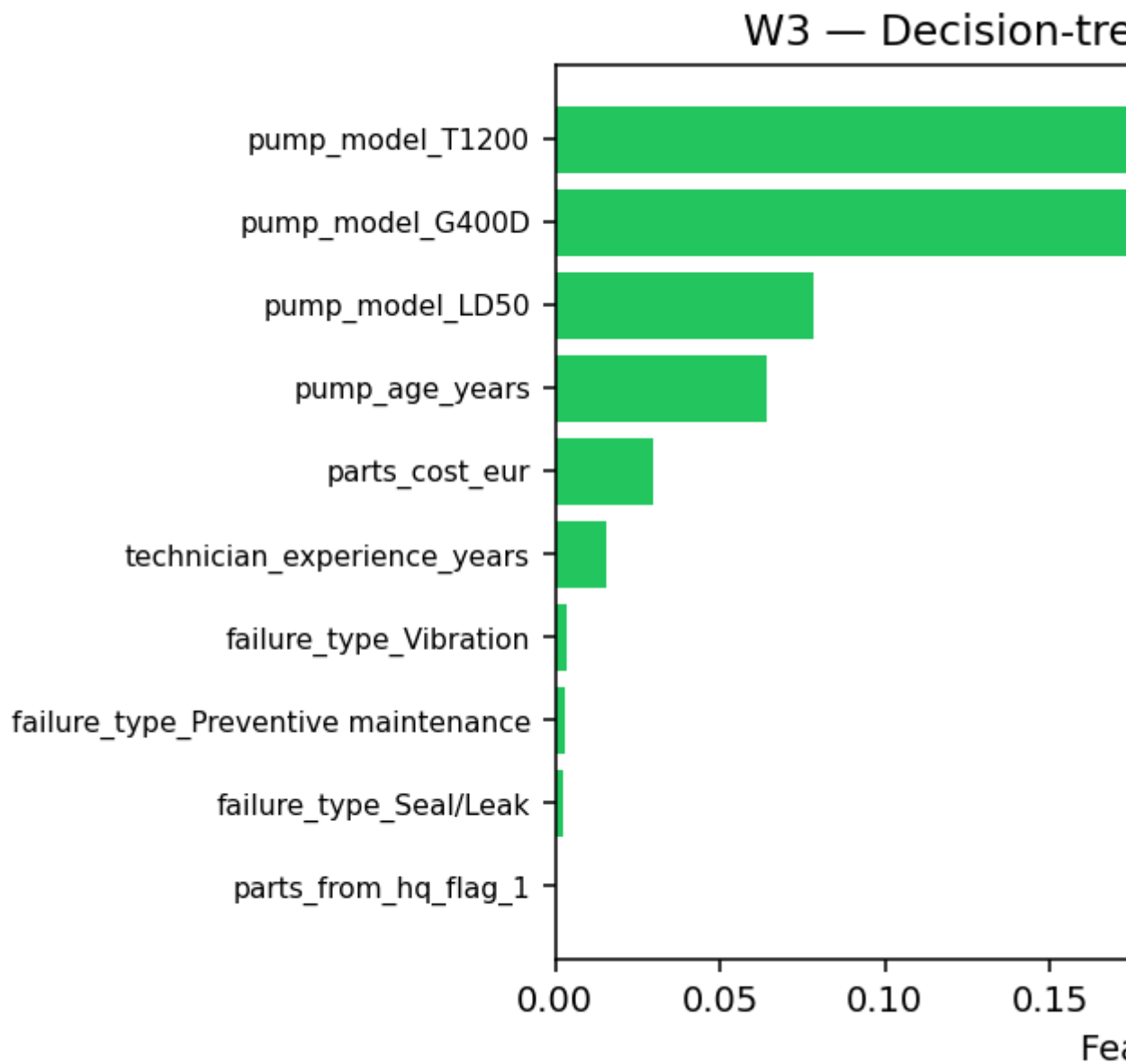
- One algorithm, one operating point — matches how a BI dashboard exposes the "flag or not" decision.
- No `class_weight`, no threshold sweep, no `predict_proba` slider.

Hyperparameters and operating point

Parameter	Value	Reason
<code>max_depth</code>	6	Same reasoning as W2
<code>min_samples</code>	2	Smooths leaves on imbalanced

Parameter	Value	Reason
es_ lea f	0	data
ran dom _st ate	7	Deterministic split (stratified on the target)
thr esh old	0 . 3 0	One sensible operating point — flags a moderate share of jobs, materially better than random





Reading the charts:

- **Confusion matrix** — at threshold 0.30 the model flags a moderate share of jobs; some failures still slip through (this is the cost of using one threshold and one simple algorithm).
- **Feature importance** — complexity, technician experience, and parts sourcing dominate — the same factors a senior QA supervisor would cite.

Management action

- Route the flagged jobs through a 5-minute senior-tech **pre-QA inspection**.

- Monitor the **flag-vs-failure ratio** weekly.
- If the workshop has spare capacity, lower the threshold to catch more failures; if not, raise it.

Conclusion / implications: A shallow Decision Tree at a fixed threshold delivers a governable pre-QA triage signal: moderate recall at a single operating point, with decision logic explainable in one sentence to operations leadership.

7. Analytical summary

All work packages share a common evaluation discipline: fixed train/test split (random_state=7), leakage-safe feature sets, and headline KPIs in exports/metrics_summary.json. The live table in §3 reflects the latest run.

Interpretation:

- **W1** — Stable co-occurrence structure in PM baskets; the Pareto shortlist is actionable without model tuning.
- **W2** — Usable intake-time ETAs from a shallow tree; residual spread reflects genuine heterogeneity at booking time.
- **W3** — Deliberate trade-off: one threshold (0.30) balances recall against inspection workload; suitable for a pre-QA queue, not maximal recall.

Together, the three models form a coherent service analytics layer — stocking, scheduling, and quality gate — each auditable in a single management review.

8. Limitations & next steps

- **Synthetic data.** The dataset is generated from a Phase-1 simulator. Numbers should be re-measured on a real production extract.
 - **W3 timing.** The model includes `repair_duration_days`, i.e. it scores *just before QA*. For a true intake-time risk score, drop that feature.
 - **No segmentation.** Both W2 and W3 are pooled across regions and tiers.
A future iteration could fit one tree per customer tier.
 - **No deployment automation.** ERP integration and scheduling hooks are out of scope; artefacts are analysis-ready for downstream engineering.
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9. Reproduction commands

```
# create a fresh venv and install dependencies

python3 -m venv .venv && source .venv/bin/activate

pip install -r requirements.txt

# regenerate exports + metrics

python3 bi_pipeline.py
```

```
# re-execute the notebooks headless
```

```
jupyter nbconvert --to notebook --execute --inplace \
```

```
notebooks/W1_Inventory.ipynb \
```

```
notebooks/W2_Repair_Duration.ipynb \
```

```
notebooks/W3_QA_Failure_Risk.ipynb
```

```
# rebuild this handbook PDF
```

```
python3 build_handbook.py
```

```
# smoke tests
```

```
python3 -m pytest tests/ -v
```

Live KPIs are injected at PDF build time from `exports/metrics_summary.json`.